Weekly Software Development

Principles of Web Design

Foundation (Basic) Learning Path

MEET 1

Agenda – What We'll Cover Today

01 Basic Elements in HTML

02 An Intro to CSS

03 The Holy Grail of Layouting

04 Border, Margin & Padding

05 Z-Index

06 Element Attributes

07 Typography & Readability

08 Colors & Contrast

01 · Basic Elements in HTML

The building blocks of every web page

HTML STRUCTURE

```
<html>  <head>  <body>
→ Struktur wajib setiap halaman
```

LINKS & IMAGES

```
<a href="...">  <img src="..." alt="...">
→ Navigasi & media visual
```

TEXT ELEMENTS

```
<h1>–<h6>  <p>  <span>  <div>
→ Heading, paragraf, inline & blok
```

LISTS & FORMS

```
<ul>  <ol>  <li>  <input>  <button>
→ Daftar & elemen interaktif form
```

Tip: Setiap elemen HTML punya peran semantik — gunakan yang tepat untuk struktur yang clean & accessible.

02 · An Intro to CSS

Styling the web — colors, fonts & the box model

Inline

`style="color:red"`

Langsung di tag HTML

Internal

`<style> ... </style>`

Di dalam <head>

External

`<link rel="stylesheet">`

File .css terpisah

SELECTORS

```
element   →  p { ... }
.class    →  .card { ... }
#id       →  #main { ... }
:hover    →  a:hover { ... }
```

color · background-color · font-family · font-size

BOX MODEL BASICS

03 · The Holy Grail of Layouting

Flexbox · CSS Grid · Table Layout

FLEXBOX

```
display: flex
flex-direction: row | column
justify-content: center | space-between
align-items: center | stretch
gap: 1rem
```

item 1 | item 2 | item 3

← flex row →

CSS GRID

```
display: grid
grid-template-columns: 1fr 2fr
grid-template-rows: auto
grid-area: header | main | aside
gap: 16px
```

header

aside | main

footer

BONUS: display: table / table-row / table-cell — fallback layout untuk browser lama, berguna untuk alignment vertikal.

03 · The Holy Grail of Layouting

Flexbox VS CSS Grid

FLEXBOX (1D)

- Layout satu arah (row atau column)
- Fokus: flow & alignment
- Konten lebih “fluid” (ikut isi)
- Cocok untuk:
  - Navbar
  - Button group
  - Form (vertical/horizontal)
  - Centering

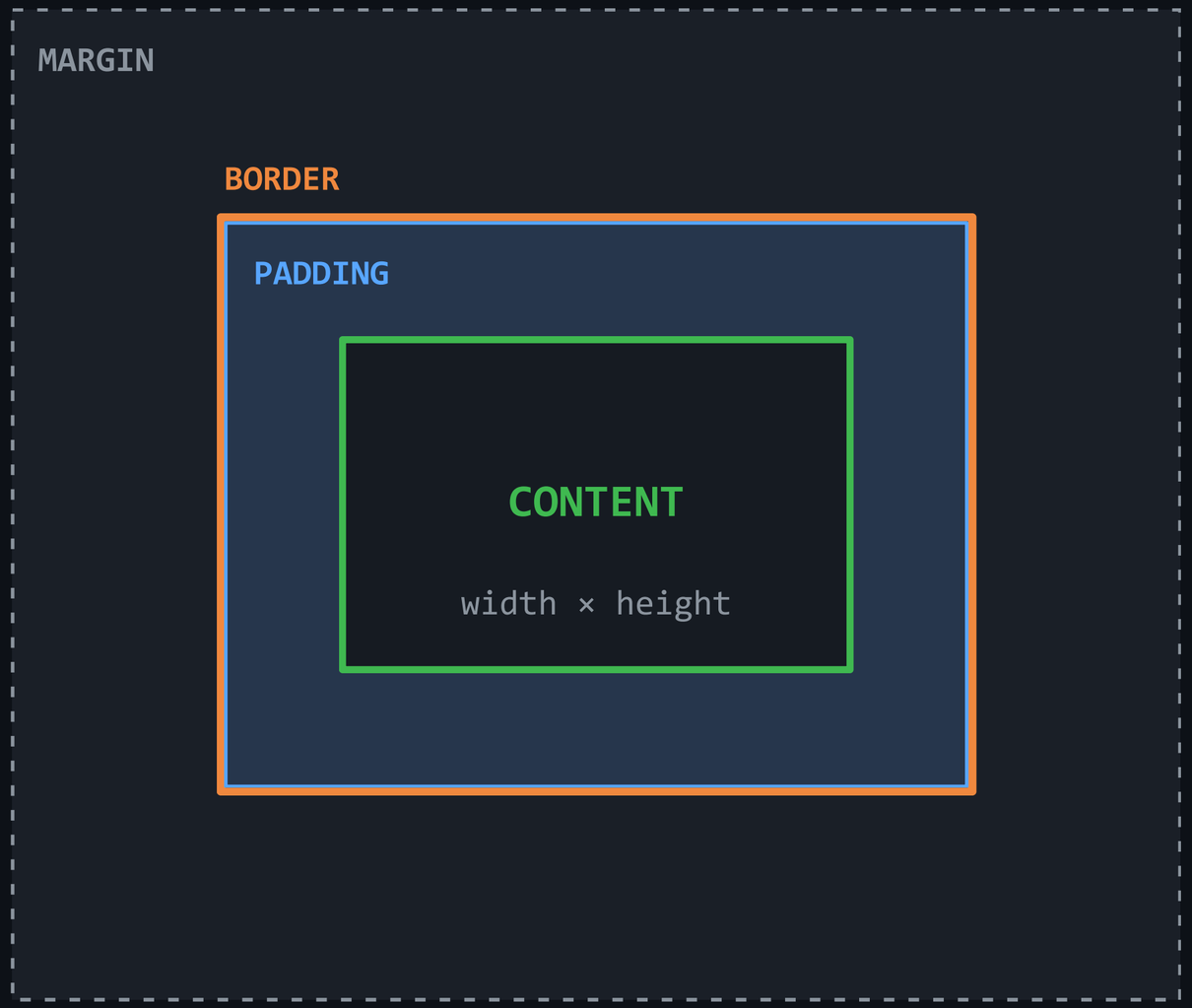
CSS GRID(2D)

- Layout dua arah (row dan column)
- Fokus: struktur & posisi
- Lebih presisi & konsisten
- Cocok untuk:
  - Card/catalog
  - Dashboard
  - Layout halaman
  - Gallery

Summary: Gunakan Flexbox untuk mengatur elemen dalam satu arah (alignment & flow), dan Grid untuk membuat layout terstruktur dua arah (row dan column) dengan kontrol yang lebih presisi.

04 · Border, Margin & Padding

Box model lebih detail — ukuran & jarak antar elemen



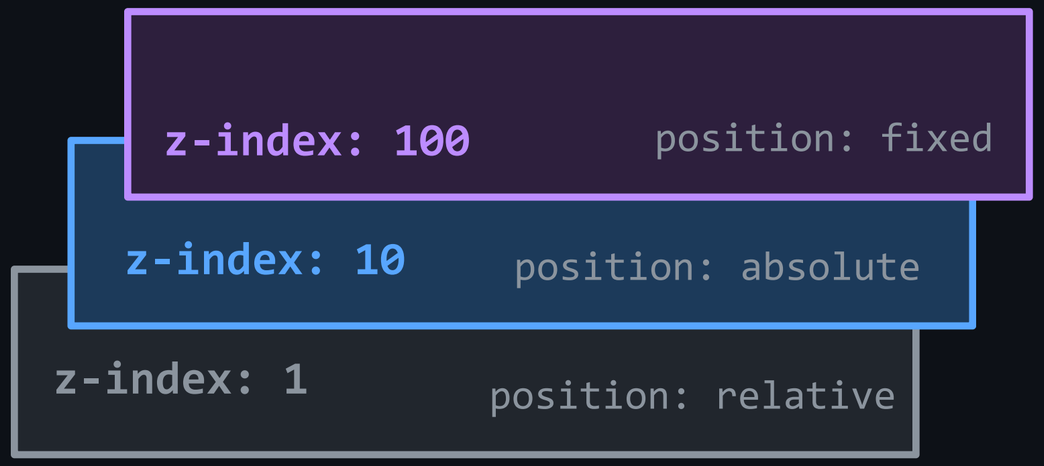
CSS PROPERTIES

margin:	top right bottom left
padding:	top right bottom left
border:	2px solid #58A6FF
width:	300px | 50% | auto
height:	200px | auto | 100vh
box-sizing:	border-box

05 · Z-Index

Layer ordering & stacking context

higher z-index = on top ↑



z-index hanya bekerja pada positioned elements

`position: relative | absolute | fixed | sticky`

Stacking context baru dibuat oleh:

`position + z-index, opacity < 1, transform, filter, dll`

z-index: auto vs z-index: 0

`auto = tidak membuat stacking context baru; 0 = membuat`

06 · Element Attributes

id, class, name, style, title & CSS selector terkait

id

`id="main-header"`

Unik per halaman · CSS #id · JS getElementById

style

`style="color:red"`

Inline CSS — hindari, gunakan class

class

`class="card active"`

Bisa banyak · CSS .class · JS classList

title

`title="Tooltip text"`

Tooltip hover & accessibility hint

name

`name="username"`

Form data key · backend reference

data-*

`data-id="42"`

Custom data attr · JS dataset API

07 · Typography & Readability

Font, spacing & visual hierarchy yang bersih

Aa

The quick
brown fox

jumps over the lazy dog

The quick brown fox

THE QUICK BROWN FOX

font-size:	16px body · 24–48px heading
font-family:	sans-serif, monospace, serif
font-weight:	400 normal · 700 bold
line-height:	1.5–1.8 untuk readability
letter-spacing:	0–0.1em (tracking)
text-align:	left (body) · center (header)

Hierarchy: H1 > H2 > body text — contrast ukuran yang jelas membantu user scan konten.

08 · Colors & Contrast

Format warna, pemilihan palet & aksesibilitas

HEX

`#58A6FF`

Paling umum, 6 digit

RGBA

`rgba(88,166,255,0.8)`

Opacity support

HSL

`hsl(212, 100%, 67%)`

Intuitif untuk variasi warna

CONTRAST EXAMPLES

Aa	Aa	Aa	Aa
White / Black	Blue / Dark	Gray / Gray	Green / Dark
AAA 21:1	AA 4.8:1	Fail 1.5:1	AA 7.2:1

WCAG 2.1: Kontras minimum AA = 4.5:1 untuk teks normal, 3:1 untuk teks besar. Gunakan tools seperti Colour Contrast Analyser atau browser DevTools untuk cek.

THAT'S A WRAP!

Keep Building, Keep Styling.

Recap:

- HTML structure & semantic elements
- CSS styling — selectors, colors, fonts
- Flexbox & Grid layouting
- Box model: margin, border, padding
- Z-index & stacking context
- Attributes, typography & contrast